535510 Reinforcement Learning

Pre-Lecture Assignment 4

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Problem: Value-Based RL

In Lectures 17-19, we proceed to introduce value-based RL methods. Let's try to get familiar with these methods and get prepared for Lectures 18-19. You can refer to the (1) attached reference slides on Value-based RL and the (2) Rainbow paper (https://arxiv.org/abs/1710.02298) as a starting point when answering the questions below.

1) Q-learning: Can you explain how the Q-learning update is derived? Please provide at least 2 different interpretations.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha_t(s_t, a_t)\Big(r_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)\Big)$$

Can you explain why Q-learning is "off-policy"?

2) Deep Q Network (DQN): The standard DQN loss function is as follows. Explain the meanings of each component in the loss. Can you explain the principle behind this design?

$$L(\mathbf{w}) := \frac{1}{2}\mathbb{E}_{(s,a,r,s')\sim\rho}\Big[\Big(r + \gamma \max_{a'\in A} \hat{Q}(s', a'; \mathbf{w}) - \hat{Q}(s, a; \mathbf{w})\Big)^2\Big]$$

3) Practical Implementation of DQN: What is "Noisy Net" in the Rainbow paper?

Submission Guidelines and Remarks

- Please write all your responses within **1 page** (**A4 paper**, either hand-written or typeset). The responses that go beyond 1 page will *not* be graded.

- Your deliverable shall be a PDF file. Please put all your write-ups into one PDF file (photos/scanned copies are acceptable) and submit your deliverable via E3.

- Please submit it by **9pm, 5/12 (Tuesday)**.